

Consuming Senders from Coroutine-Native Code

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Abstract

An `IoAwaitable` bridge ([P4003R0](#)^[3]) consumes `std::execution` senders with inline operation state, correct stop token propagation, and automatic executor dispatch-back. The bridge is one class template. The complete implementation is in Appendix A.

Revision History

R0: March 2026 (post-Croydon mailing)

- Initial version.
-

1. Disclosure

The authors developed and maintain [Capy](#)^[4] and [Corosio](#)^[6] and believe coroutine-native I/O is the correct foundation for networking in C++. The findings in this paper are structural and hold regardless of which library implements the coroutine-native layer. This paper is one of a suite of four that examines the relationship between compound I/O results and the sender three-channel model. The companion papers are [D4053R0](#)^[2], “Sender I/O: A Constructed Comparison”; [D4054R0](#)^[11], “Two Error Models”; and [D4056R0](#)^[12], “Producing Senders from Coroutine-Native Code.” The authors provide information, ask nothing, and serve at the pleasure of the chair. [Capy](#)^[4] is a coroutine primitives library. It contains no sockets, no timers, and no platform-specific I/O APIs. Networking is in [Corosio](#)^[6], which is not used in this paper. The bridge depends on [Capy](#)^[4] and [beman::execution](#)^[5], a community implementation of [std::execution](#) ([P2300R10](#)^[1]). Source code links refer to pinned commit [f046646](#)^[4].

2. The Bridge

```
copy::task<int> compute(auto sched)
{
    int result = co_await copy::await_sender(
        ex::schedule(sched)
        | ex::then([] {
            std::cout
                << " sender running on thread "
                << std::this_thread::get_id()
                << "\n";
            return 42 * 42;
        }));

    std::cout
        << " coroutine resumed on thread "
        << std::this_thread::get_id() << "\n";

    co_return result;
}
```

`await_sender` returns a `sender_awaitable` that satisfies `IoAwaitable` (P4003R0^[3]). Any coroutine type whose promise propagates `io_env` through `await_suspend(h, io_env const*)` can use it. `copy::task` is one such type. It is not the only one.

Appendix A contains the complete bridge implementation. It was tested against `beman::execution` ^[5].

3. Demonstration

```
main thread: 32208
sender running on thread 9560
coroutine resumed on thread 34356
result: 1764
```

The sender executed on thread 9560, the `beman::execution::run_loop` thread. The coroutine resumed on thread 34356, the Capy thread pool. The value 1764 crossed the bridge. The bridge allocated zero bytes on this path.

4. What the Bridge Does

The bridge consumes any `std::execution` sender. Scheduler hops work. Stop token propagation works. `set_value`, `set_error`, and `set_stopped` are handled. The operation state is stored inline. Any sender pipeline - `when_all`, `then`, `let_value`, `on` - works through the bridge.

An `IoAwaitable` coroutine can suspend into a sender pipeline that schedules work on a GPU, a thread pool, or any other execution context. When the sender completes, the bridge dispatches the resumption back to the coroutine's originating executor (`env_->executor.post(cont_)`). The coroutine resumes in the correct context with the sender's result, regardless of where the sender executed. The bridge is the interop path for coroutine-native I/O code that needs to offload compute to a GPU scheduler and resume on the I/O event loop.

The bridge does not use `execution::task`.

5. What the Bridge Does Not Require

Property	<code>execution::task</code>	Bridge
Routine I/O errors become exceptions	Yes	No
Type erasure on connect	Yes	No
<code>dispatch</code> adapter for compound results	Yes	No
Zero allocations in the bridge	No	Yes

`std::execution::task` is not necessary to consume senders.

6. The Narrowest Abstraction

`Capy`^[4] is a coroutine primitives library. It provides `task`, executors, stop tokens, frame allocators, and buffer sequences. It contains no sockets, no timers, and no platform-specific I/O APIs. Networking is in `Corosio`^[6], a separate library that depends on `Capy`^[4].

The sender bridge depends on `Capy`^[4] and `std::execution`. It does not depend on `Corosio`^[6]. It does not depend on any platform I/O API. A user who needs I/O adds `Corosio`^[6]. A user who needs sender interop adds the bridge. A user who needs neither does not pay for either. The coroutine type does not change between these configurations.

7. Acknowledgments

The authors thank Dietmar Kühl for `beman::execution`^[5] and for the channel-routing enumeration in `P2762R2`^[7], Michał Dominiak, Eric Niebler, and Lewis Baker for `std::execution`, Chris Kohlhoff for identifying the partial-success problem in `P2430R0`^[8], Kirk Shoop for the completion-token heuristic analysis in `P2471R1`^[9], Fabio Fracassi for `P3570R2`^[10], Ville Voutilainen and Jens Maurer for reflector discussion on dispatch patterns, Herb Sutter for identifying the need for tutorials and documentation, Mark Hoemmen for insights on `std::linalg` and the layered abstraction model, and Peter Dimov for the refined channel mapping.

References

1. `P2300R10` - “std::execution” (Michał Dominiak et al., 2024). <https://wg21.link/p2300r10>
2. `D4053R0` - “Sender I/O: A Constructed Comparison” (Vinnie Falco, Steve Gerbino, 2026). <https://wg21.link/d4053r0>
3. `P4003R0` - “Coroutines for I/O” (Vinnie Falco, Steve Gerbino, Mungo Gill, 2026). <https://wg21.link/p4003r0>
4. `cppalliance/capy` - Coroutine primitives library. Commit f046646. <https://github.com/cppalliance/capy/tree/f0466466e63baf0cc3d6034bc35eec24694f5d16>
5. `bemanproject/execution` - Community implementation of `std::execution`. <https://github.com/bemanproject/execution>
6. `cppalliance/corosio` - Coroutine-native networking library. <https://github.com/cppalliance/corosio>
7. `P2762R2` - “Sender/Receiver Interface For Networking” (Dietmar Kühl, 2023). <https://wg21.link/p2762r2>
8. `P2430R0` - “Partial success scenarios with P2300” (Chris Kohlhoff, 2021). <https://wg21.link/p2430r0>
9. `P2471R1` - “NetTS, ASIO and Sender Library Design Comparison” (Kirk Shoop, 2021). <https://wg21.link/p2471r1>
10. `P3570R2` - “Optional variants in sender/receiver” (Fabio Fracassi, 2025). <https://wg21.link/p3570r2>

11. **D4054R0** - "Two Error Models" (Vinnie Falco, 2026). <https://wg21.link/d4054r0>

12. **D4056R0** - "Producing Senders from Coroutine-Native Code" (Vinnie Falco, Steve Gerbino, 2026).
<https://wg21.link/d4056r0>

Appendix A. Bridge Implementation

```
#include <boost/capy/ex/io_env.hpp>

#include <beman/execution/execution.hpp>

#include <cassert>
#include <coroutine>
#include <cstring>
#include <exception>
#include <new>
#include <stop_token>
#include <stdexcept>
#include <tuple>
#include <type_traits>
#include <utility>
#include <variant>

namespace boost::capy {

namespace detail {

struct stopped_t {};

struct bridge_env
{
    std::stop_token st_;

    auto query(
        beman::execution::get_stop_token_t const&)
        const noexcept
    {
        return st_;
    }
};

template<class Sender>
using sender_single_value_t =
    beman::execution::value_types_of_t<
        Sender,
        bridge_env,
        std::tuple,
        std::type_identity_t>;

template<class ValueTuple>
```

```

struct bridge_receiver
{
    using receiver_concept =
        beman::execution::receiver_t;

    std::variant<
        std::monostate,
        ValueTuple,
        std::exception_ptr,
        stopped_t>*      result_;
    std::coroutine_handle<> cont_;
    io_env const*        env_;

    auto get_env() const noexcept -> bridge_env
    {
        return {env_ -> stop_token};
    }

    template<class... Args>
    void set_value(Args&&... args) && noexcept
    {
        result_ -> template emplace<1>(
            std::forward<Args>(args)...);
        env_ -> executor.post(cont_);
    }

    template<class E>
    void set_error(E&& e) && noexcept
    {
        if constexpr (
            std::is_same_v<
                std::decay_t<E>,
                std::exception_ptr>)
            result_ -> template emplace<2>(
                std::forward<E>(e));
        else
            result_ -> template emplace<2>(
                std::make_exception_ptr(
                    std::forward<E>(e)));
        env_ -> executor.post(cont_);
    }

    void set_stopped() && noexcept
    {
        result_ -> template emplace<3>(
            stopped_t{});
        env_ -> executor.post(cont_);
    }

```



```

    }
};

} // namespace detail

template<class Sender>
struct sender_awaitable
{
    using value_tuple =
        detail::sender_single_value_t<Sender>;
    using receiver_type =
        detail::bridge_receiver<value_tuple>;
    using op_state_type = decltype(
        beman::execution::connect(
            std::declval<Sender>(),
            std::declval<receiver_type>()));

    Sender sndr_;

    std::variant<
        std::monostate,
        value_tuple,
        std::exception_ptr,
        detail::stopped_t> result_{};

    alignas(op_state_type)
    unsigned char op_buf_[sizeof(op_state_type)];
    bool op_constructed_ = false;

    explicit sender_awaitable(Sender sndr)
        : sndr_(std::move(sndr))
    {
    }

    sender_awaitable(sender_awaitable&& o)
        noexcept(
            std::is_nothrow_move_constructible_v<
                Sender>)
        : sndr_(std::move(o.sndr_))
    {
        assert(!o.op_constructed_);
    }

    sender_awaitable(
        sender_awaitable const&) = delete;
    sender_awaitable& operator=(
        sender_awaitable const&) = delete;

```

```

sender_awaitable& operator=(
    sender_awaitable&&) = delete;

~sender_awaitable()
{
    if(op_constructed_)
        std::launder(
            reinterpret_cast<op_state_type*>(
                op_buf_))->~op_state_type();
}

bool await_ready() const noexcept
{
    return false;
}

std::coroutine_handle<>
await_suspend(
    std::coroutine_handle<> h,
    io_env const* env)
{
    ::new(op_buf_) op_state_type(
        beman::execution::connect(
            std::move(sndr_),
            receiver_type{
                &result_, h, env}));
    op_constructed_ = true;
    beman::execution::start(
        *std::launder(
            reinterpret_cast<
                op_state_type*>(
                    op_buf_)));
    return std::noop_coroutine();
}

auto await_resume()
{
    if(result_.index() == 2)
        std::rethrow_exception(
            std::get<2>(result_));
    if(result_.index() == 3)
        throw std::runtime_error(
            "sender completed with "
            "set_stopped");

    if constexpr (
        std::tuple_size_v<

```

```

        value_tuple> == 0)
    return;
    else if constexpr (
        std::tuple_size_v<
            value_tuple> == 1)
        return std::get<0>(<
            std::get<1>(<
                std::move(result_))));
    else
        return std::get<1>(<
            std::move(result_));
    }
};

template<class Sender>
auto await_sender(Sender&& sndr)
{
    return sender_awaitable<
        std::decay_t<Sender>>(<
            std::forward<Sender>(sndr)));
}

} // namespace boost::copy

```